Course Name - Exercise 1

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Question 1

(a)

Proof. Let X and Y be two independent random variables (assuming they are continues).
By definition: $f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$. therefor:

$$E[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf_{X,Y}(x,y)\,dx\ dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} yf_Y(y)\,(xf_X(x))\,dx\ dy =$$

$$\int_{-\infty}^{\infty} yf_Y(y) \int_{-\infty}^{\infty} xf_X(x)dx\ dy = \int_{-\infty}^{\infty} yf_Y(y)E[X]\,dy = E[X] \int_{-\infty}^{\infty} yf_Y(y)dy = E[X]\,E[Y]$$

4
As required. ∎

(b)

Let X, Y be two RV such that $X \sim U(-1,1), Y = X^2$. Obviously the RV are dependent. Lets show they are uncorrelated:

$$E[X] = 0, E[Y] = E\left[X^2\right] = \int_{-1}^{1} x^2 \cdot \frac{1}{2}dx = \frac{1}{2}\left(\frac{1}{3} - \left(-\frac{1}{3}\right)\right) = \frac{1}{3}$$

$$E[XY] = E\left[X^3\right] = \int_{-1}^{1} x^3 \cdot \frac{1}{2}dx = \frac{1}{2}\left(\frac{1}{4} - \frac{1}{4}\right) = 0$$

We get $E[XY] = E[X]\,E[Y]$ thus the RV are uncorrelated as required.

(c)

Let X, Y be two discrete RV such that $X, Y \sim U\{0,1\}$ and let Z be their symmetric difference $Z = X \oplus Y$.
Of course X, Y are independent and knowing any two of the RV fixes the last one. Lets show X, Z are independent (symmetric proof for Y, Z):
The original probabilities for Z to be 0 or 1 is 50% ($Z = 1$: either $X = 1$ and $Y = 0$ or vice verse, $Z = 0$: either both equal 1 or 0).
Now, let's assume we know the value of X. The value of Z is still 0 or 1 with 50% for each value because it is dependent only on the value of Y.
On the other hand, assume we now the value of Z. It won't affect the value of X because it is still a uniform random variable, and for any value of X, Y can be set in such a way that $X \oplus Y = Z$.

(d)

Proof. let $x_1, ..., x_n, C$ be as needed. Let $z \in \mathbb{R}^d$.

$$z^T C z = z^T \sum_{i=1}^{n} x_i x_i^T z = \sum_{i=1}^{n} z^T x_i x_i^T z = \sum_{i=1}^{n} \left(z^T x_i\right) \cdot \left(z^T x_i\right)^T$$

Notice each number in the sum is a real number raised to the second power, which is always greater or equal 0, as required. ■

Question 2

(a)

Assuming the one- dimensional case (d=1), we are in a 1 variable normal distribution, so $X \sim N(0, 1)$.
We want to calculate $P(1 > X > -1)$. Using the normal distribution table, we get:

$$P(1 > X > -1) = 2(P(X < 0) - P(X < -1)) \approx 2(0.5 - 0.15866) = 0.6827$$

As needed.

(b)

Proof. The probability of each RV x_i in the vector to be bounded between $-1 \leftrightarrow 1$ is around 0.6827 (as described above). For any dimension, we can inscribe the unit sphere inside a cube that is bounded by $(1, 1, 1, ..., 1)$ and $(-1, -1, -1, ..., -1)$. The probability of a vector being inside this cube is equivalent to the probability of each RV in the vector to be bounded between $-1 \leftrightarrow 1$:

$$\lim_{d\to\infty}\left(\prod_{i=1}^{d} P(-1 < x_i < 1)\right) = \lim_{d\to\infty}\left(\prod_{i=1}^{d} 0.6827\right) = 0$$

The sphere is inscribed inside the cube and therefor the probability of the vector lending inside the sphere is even lower. ∎

Question 3

Proof. From the definition of variance:

$$Var\left(y\right)=E\left[\left(Y-E\left[Y\right]\right)\left(Y-E\left[Y\right]\right)^{T}\right]=E\left[\left(w^{T}X-E\left[w^{T}X\right]\right)\left(w^{T}X-E\left[w^{T}X\right]\right)^{T}\right]=$$

$$E\left[\left(w^{T}\left(X-E\left[X\right]\right)\right)\left(w^{T}\left(X-E\left[X\right]\right)\right)^{T}\right]=w^{T}E\left[\left(X-E\left[X\right]\right)\left(X-E\left[X\right]\right)^{T}\right]w=w^{T}Cw$$

■

Question 4

(a)

Proof. A has n eigenvalues, therefor is decomposition-able: $A = PUP^{-1}$ where U is a jordan matrix with A's eigenvalues in it's diagonal.

$$A^T = \left(PUP^{-1}\right)^T \overset{(AB)^T = B^T A^T}{=} {P^{-1}}^T U^T P^T \overset{{A^T}^{-1} = {A^{-1}}^T}{=} {P^T}^{-1} U^T P^T$$

We found a eigendecomposition for A^T ${P^T}^{-1} U P^T$ where P^T is invertible and U^T is in Jordan form. It means the eigenvalues of A^T are the values on the diagonal of U^T which are the same values on the diagonal of U as required. ■

(b)

Proof. A is diagonalizable: $A = PUP^{-1}$. Therefor:

$$\det(A) = \det\left(PUP^{-1}\right) = \det(P)\det(U)\det\left(P^{-1}\right) = \det(U)\det\left(PP^{-1}\right) = \prod_i \lambda_i \cdot 1$$

As required. ■

Question 5

(a) $f(x) = \begin{cases} 0 & w^T x \geq 1 \\ 1 - w^T x & w^T x < 1 \end{cases}$

Lets differentiate the function bellow 1 and above 1:

$$w^T x > 1 : \nabla f(x) = 0. \; w^T x < 1 : \nabla f(x) = -w$$

Assuming $w \neq 0$, the function isn't differentiable at $x = 1$.

(b) $f(x) = \frac{1}{1+e^{-w^T x}}$

The function is a composition of elementary functions, therefor is differentiable for all x (notice $e^x \neq -1$ for any rational x).

$$\nabla f(x) = -1 \cdot \frac{e^{-w^T x} \cdot (-w)}{\left(1 + e^{-w^T x}\right)^2} = \frac{1}{1 + e^{-w^T x}} \cdot \left(\frac{1 + e^{-w^T x}}{1 + e^{-w^T x}} - \frac{1}{1 + e^{-w^T x}}\right) \cdot w = f(x)\left(1 - f(x)\right) w$$

As required.